

# Smart Traffic Signal Control Using Local AI: A Sample Architecture for Emergency Vehicle Route Clearance

K. CHARAN REDDY

USN: 3BR25CD047

Department of CSE-DS

Ballari, India

Email: charanreddy2006k@gmail.com

**Abstract**—Traditional traffic signals commonly operate using predefined timing plans. Such approaches are simple to implement but may not respond effectively to rapidly changing traffic conditions. Recent research has investigated artificial intelligence and machine-vision techniques for adaptive traffic signal control.

This paper presents a sample architecture for a local-AI-assisted traffic signal system. The proposed architecture uses street cameras to obtain traffic information, local computing units to process the information, and traffic-light controllers to adjust signal operation. A cloud server may be used for storage and large-scale analysis rather than requiring all processing to occur in the cloud. The paper also considers emergency vehicle route clearance as a possible application of the architecture. The proposed system is presented as a design concept and does not claim experimental performance results.

**Index Terms**—Traffic Signal Control, Artificial Intelligence, Machine Vision, Local AI, Emergency Vehicles, Smart Transportation

## I. INTRODUCTION

Traffic congestion is a common problem in urban transportation systems. Conventional traffic lights generally operate according to predefined timing schedules. Although fixed timing is simple and inexpensive, it may not react efficiently when traffic conditions change significantly.

Artificial intelligence (AI) has been investigated as a method for adaptive traffic signal control. A review by Rasheed *et al.* discusses the use of deep reinforcement learning for traffic signal control and summarizes research directions in this area [1].

Machine-vision approaches have also been investigated for adaptive traffic-light control. Gorodokin *et al.* studied the optimization of adaptive traffic-light control modes using machine vision [2].

Based on these research directions, this paper presents a sample architecture in which traffic information is processed locally near the road. The main objective is to describe how local processing, cloud support, and traffic-light control can be combined in a single system.

An additional application considered in this paper is emergency vehicle route clearance. Instead of treating this feature as an experimentally validated result, it is presented as a

proposed extension that would require appropriate vehicle detection, communication, safety rules, and field testing.

## II. PROBLEM STATEMENT

Fixed-time traffic signals may not respond to sudden changes in traffic density. In addition, an emergency vehicle such as an ambulance may require a coordinated route through multiple intersections.

Therefore, a smart traffic signal architecture is required that can:

- observe traffic conditions,
- process traffic information,
- support adaptive signal decisions,
- reduce unnecessary dependence on continuous cloud processing, and
- provide a framework that can support emergency vehicle route clearance.

## III. OBJECTIVES

The objectives of the proposed architecture are:

- 1) To monitor traffic conditions using cameras.
- 2) To process traffic information using local computing devices.
- 3) To support adaptive traffic signal decisions.
- 4) To reduce the amount of information that must be continuously transmitted to a remote server.
- 5) To provide a possible framework for emergency vehicle route coordination.
- 6) To provide a scalable architecture that can use cloud services for storage and analysis.

## IV. LITERATURE REVIEW

Research on intelligent traffic signal control has expanded with the development of machine learning and computer vision.

Rasheed *et al.* reviewed deep reinforcement learning approaches for traffic signal control and discussed the potential of learning-based methods for traffic management [1].

Gorodokin *et al.* investigated adaptive traffic-light control using machine vision. Their work demonstrates the relevance

of visual traffic information in adaptive signal-control systems [2].

These studies provide a foundation for considering traffic information as an input to intelligent signal-control systems. However, the specific local-AI architecture and emergency route coordination presented in this paper are proposed system-design concepts rather than experimentally validated results.

## V. EXISTING SYSTEM

A basic traffic signal system can operate using fixed timing intervals. The signal changes according to a predefined schedule regardless of the current traffic condition.

The main advantages of such systems are simplicity and relatively low implementation complexity. However, a fixed schedule does not directly use real-time traffic information.

Another approach is to use centralized or cloud-based processing. In such systems, camera information can be transmitted to a remote server where processing is performed. This approach can provide access to greater computational resources, but communication requirements and network delay must be considered.

## VI. PROPOSED SYSTEM

The proposed architecture places a local computing unit near the traffic intersection. The local unit receives information from street cameras and performs traffic analysis.

The proposed system consists of four major stages:

- 1) Street cameras capture traffic information.
- 2) A local computing unit processes the captured information.
- 3) Selected information can be forwarded to a cloud server for storage and wider analysis.
- 4) Traffic-light controllers receive control instructions.

The overall information flow is:

**Street Cameras** → **Local Computing Unit** → **Cloud Server** → **Traffic-Light Controller**

The cloud server is optional for real-time local decisions. This architecture allows local processing to remain close to the intersection while cloud infrastructure can support storage and analysis.

## VII. SYSTEM ARCHITECTURE

The proposed architecture contains the following components:

- **Street Cameras:** Capture visual information from the road.
- **Local Computing Unit:** Performs local processing of traffic information.
- **Cloud Server:** Provides optional storage and large-scale analysis.
- **Traffic Controller:** Applies the generated signal-control decision.

- **Emergency Vehicle Module:** Represents a proposed mechanism for detecting and coordinating emergency vehicle movement.

Proposed architecture for local-AI-assisted traffic signal control.

## VIII. SYSTEM FLOWCHART

The basic operation of the proposed system is represented in Fig. VIII.

Flowchart of the proposed traffic signal control architecture.

## IX. MATHEMATICAL MODEL

Let the traffic information obtained from the camera be represented by:

$$X = \{x_1, x_2, \dots, x_n\} \quad (1)$$

where each  $x_i$  represents an observed traffic feature.

A local processing function can be represented as:

$$Y = f(X) \quad (2)$$

where  $Y$  represents the output of the local processing stage.

For multiple traffic lanes, a simple traffic-state matrix can be represented as:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \quad (3)$$

The matrix is a conceptual representation of traffic information; the actual features would depend on the selected detection and control algorithm.

## X. SYSTEM MODULES

Table I presents the major modules of the proposed architecture.

TABLE I  
MAJOR SYSTEM MODULES

| Module             | Function                                                       |
|--------------------|----------------------------------------------------------------|
| Camera             | Captures traffic information from the road.                    |
| Local AI Unit      | Processes traffic information near the intersection.           |
| Cloud Server       | Stores information and supports wider analysis.                |
| Traffic Controller | Receives the control decision.                                 |
| Signal             | Changes the traffic signal according to the controller.        |
| Emergency Module   | Represents the proposed emergency-route coordination function. |

TABLE II  
QUALITATIVE COMPARISON OF TRAFFIC SIGNAL APPROACHES

| Feature             | Fixed-Time        | Local AI |
|---------------------|-------------------|----------|
| Traffic Data        | Not directly used | Used     |
| Adaptation          | Limited           | Possible |
| Local Processing    | No                | Yes      |
| Cloud Dependency    | Not required      | Optional |
| Emergency Extension | Limited           | Possible |

## XI. SYSTEM COMPARISON

The following table provides a qualitative comparison of common traffic signal approaches.

This table describes the intended architecture and does not represent measured experimental performance.

## XII. IMPLEMENTATION CONSIDERATIONS

A practical implementation would require suitable cameras, a local computing platform, a traffic-signal controller interface, and software for traffic analysis.

A computer-vision model could be used to extract traffic information from camera frames. The local computing unit could then convert this information into a traffic state and pass the state to a signal-control algorithm.

For emergency vehicle support, the system would additionally need a reliable mechanism for identifying an authorized emergency vehicle. The identification mechanism should be tested carefully before being used in a real traffic environment.

## XIII. SECURITY AND SAFETY CONSIDERATIONS

Traffic signal control is a safety-critical application. Therefore, incorrect detection or communication failure should not result in unsafe signal operation.

The proposed system should include:

- fail-safe signal operation,
- authentication of control messages,
- protection against unauthorized access,
- monitoring of communication failures,
- manual override facilities, and
- extensive testing before real-world deployment.

## XIV. EXPECTED BENEFITS

The proposed architecture is intended to provide the following potential benefits:

- Use of real-time traffic information.
- Local processing close to the traffic intersection.
- Reduced need to transmit all camera information to a remote server.
- Support for adaptive traffic signal control.
- A framework for future emergency vehicle coordination.

These are expected or architectural benefits; they should not be interpreted as experimentally demonstrated results.

## XV. LIMITATIONS

The proposed architecture has several limitations.

First, camera-based systems can be affected by lighting, weather, camera position, and image quality. Second, local processing requires suitable computing hardware. Third, an emergency vehicle detection mechanism must be reliable before it can safely control traffic signals.

In addition, real-world deployment requires testing, traffic authority approval, safety validation, and integration with existing traffic-control infrastructure.

## XVI. FUTURE WORK

Future work can focus on developing and testing a complete computer-vision model for vehicle detection and traffic-density estimation.

Further work can also investigate adaptive signal-control algorithms, emergency vehicle identification, communication between intersections, and secure coordination of multiple traffic signals.

Field experiments could then be performed to measure processing time, detection accuracy, traffic waiting time, communication requirements, and system reliability.

## XVII. CONCLUSION

This paper presented a sample architecture for local-AI-assisted traffic signal control. The architecture combines street cameras, local processing, optional cloud services, and traffic-light control.

Previous research has investigated AI-based traffic signal control and machine-vision-based adaptive traffic lights [1], [2]. Building on these research directions, the proposed architecture places local processing near the traffic intersection and considers emergency vehicle route coordination as a future application.

The architecture is a design proposal rather than a claim of measured real-world performance. Future implementation and experimental evaluation are required to determine its actual effectiveness.

## REFERENCES

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